

Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?

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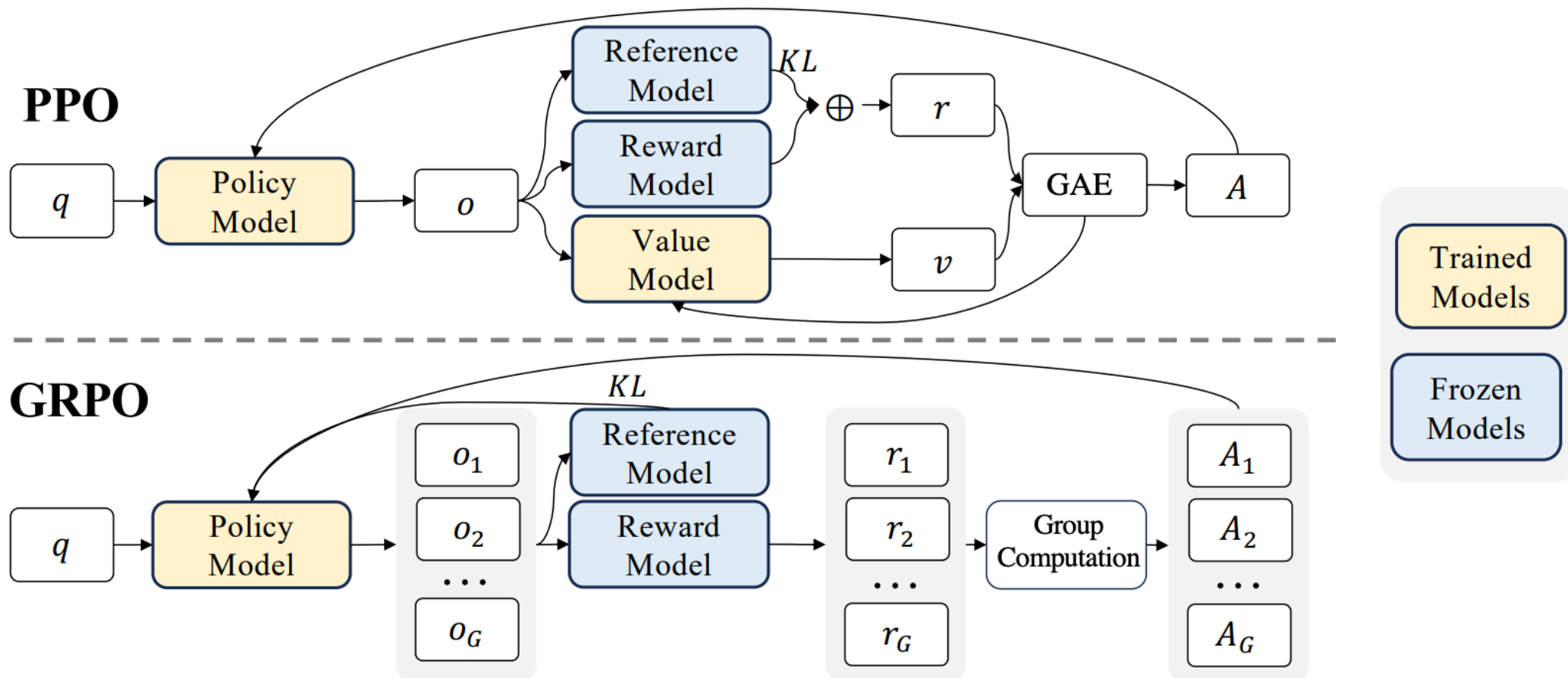
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Background

- Context:** RLVR is widely used to boost LLM reasoning (math, coding) via verifiable rewards (e.g., correct answers).
- Common Belief:** RLVR enables LLMs to learn novel reasoning patterns beyond their base models.
- Core Question:** Does current RLVR truly unlock new reasoning capabilities, or only leverage existing patterns in base models?

Background

RLVR example:



Evaluation Metric: pass@k

Question: How to evaluate a model's true reasoning potential.

Pass@k: A problem is solved if any of k sampled outputs is correct.

(Reflects the *scope of solvable problems* (reasoning boundary) rather than average performance.)

Experimental Setup

Tasks:

1. Math (GSM8K, MATH500, AIME24),
2. Code (LiveCodeBench, HumanEval+),
3. Visual Reasoning (MathVista)

Models:

1. Qwen2.5 (7B/14B/32B),
2. LLaMA-3.1-8B,
3. DeepSeek-R1,

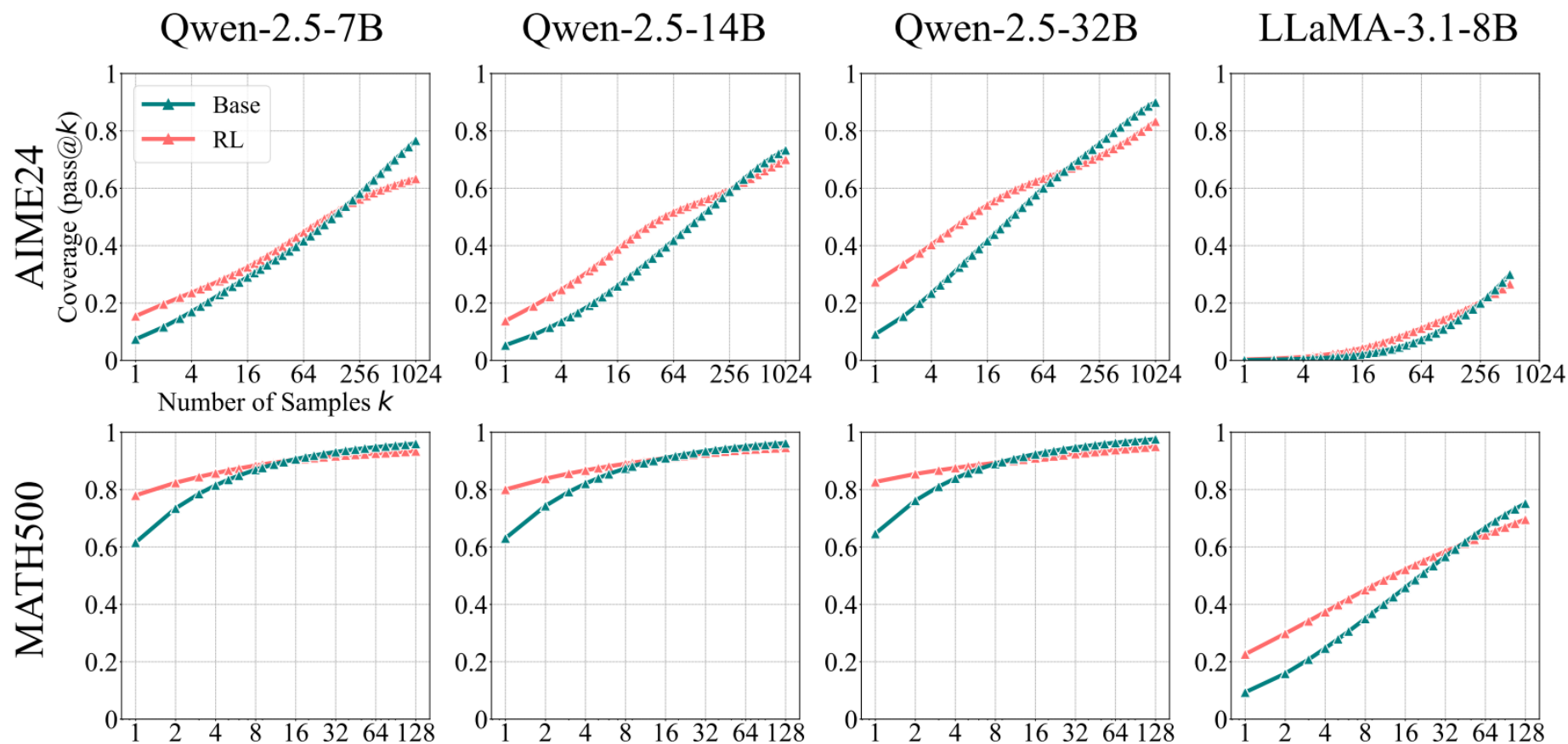
RL Algorithms:

PPO, GRPO, Reinforce++, RLOO,
ReMax, DAPO

Control:

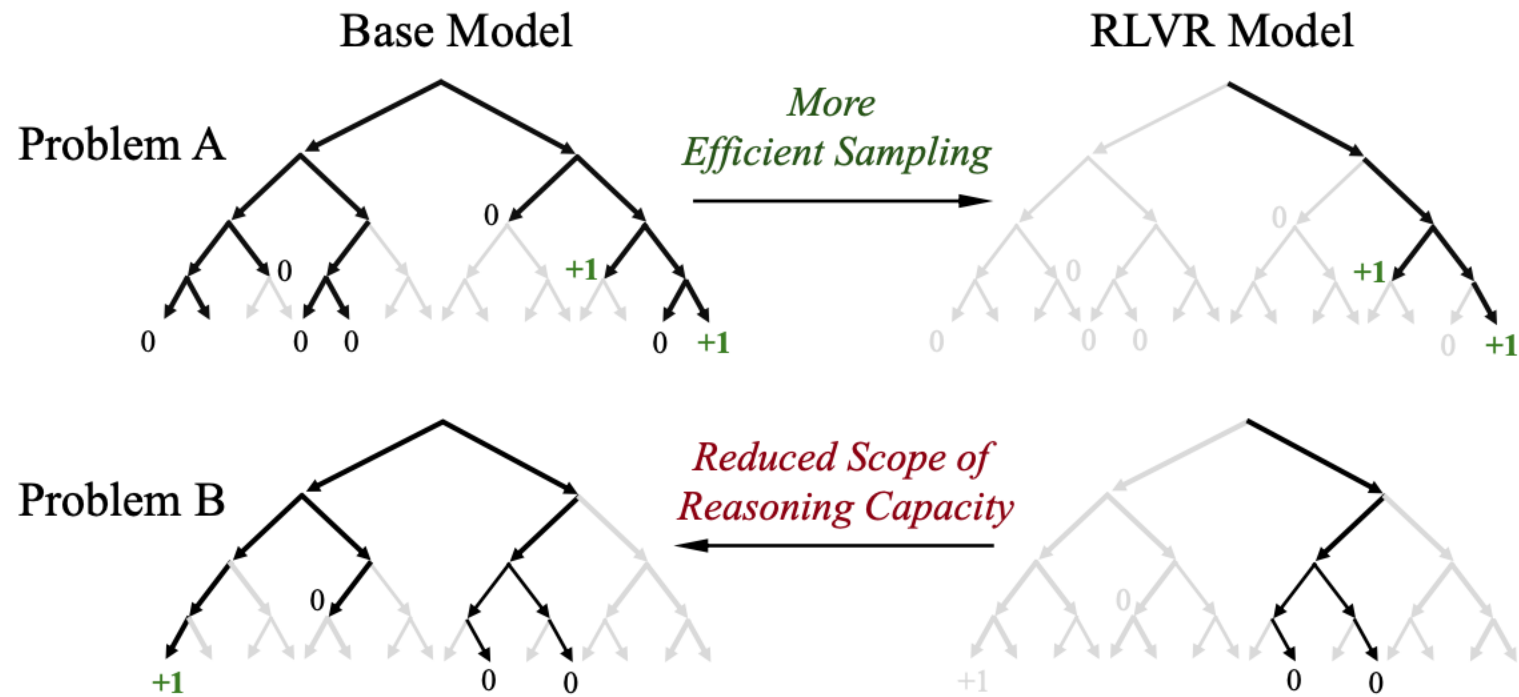
1. Zero-RL setting (no SFT) for math.
2. Instruction-tuned starts for
coding/visual tasks

Key findings



- **Small k ($k=1$):** RLVR models outperform base models (better sampling efficiency).
- **Large k ($k=128/256$):** Base models surpass RLVR models across all tasks/benchmarks.

Key findings



RLVR Increased Likelihood of Correct Samples, Decreased Coverage of Solvable Problems.

Key findings

Accuracy Distribution Analysis

1. Increases the frequency of high accuracies near 1.0 and reduces the frequency of low accuracies.
2. Increased frequency at accuracy 0: leads to more unsolvable problems.

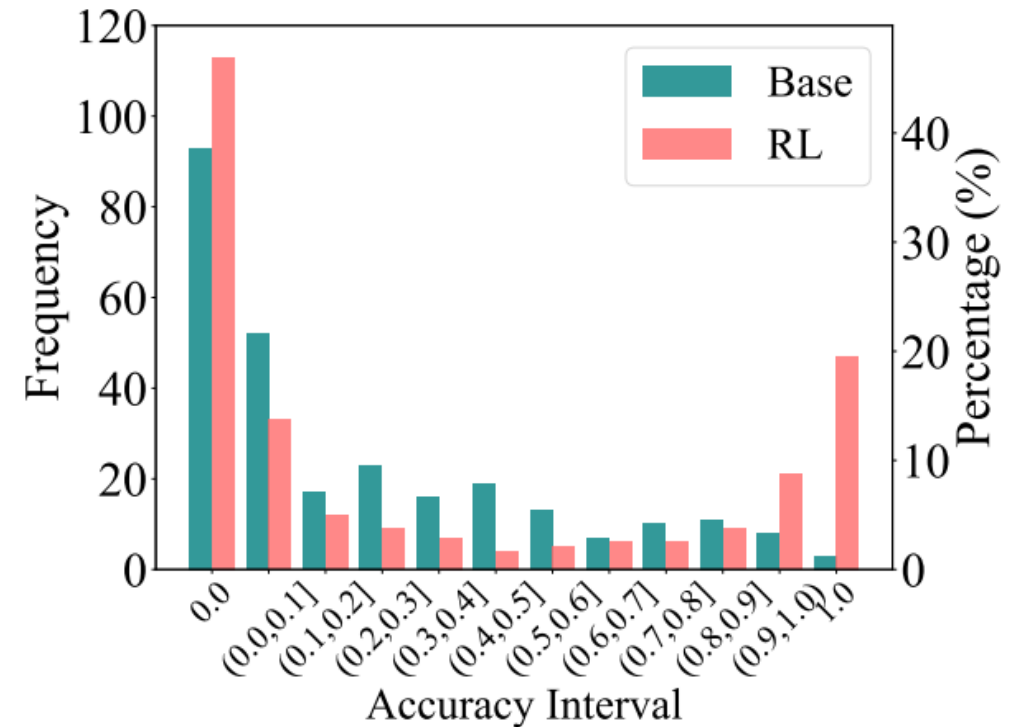


Figure 5: Qwen2.5-7B Accuracy Histogram on Minerva.

Key findings

Solvable-Problem Coverage Analysis.

Table 2: We evaluate on AIME24 ($k = 1024$) and MATH500 ($k = 128$). The table reports the solvable/un-solvable fraction of problems falling into four categories.

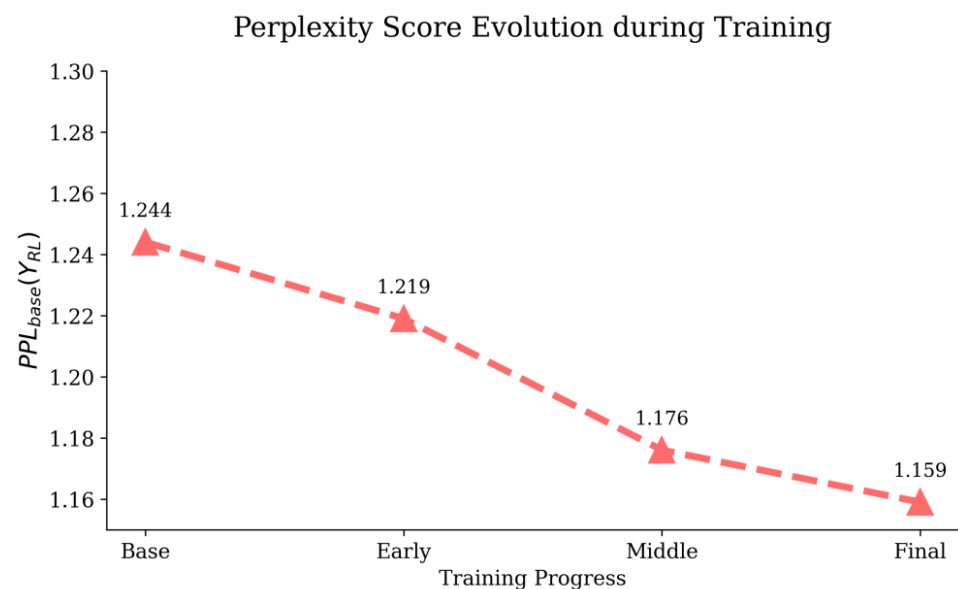
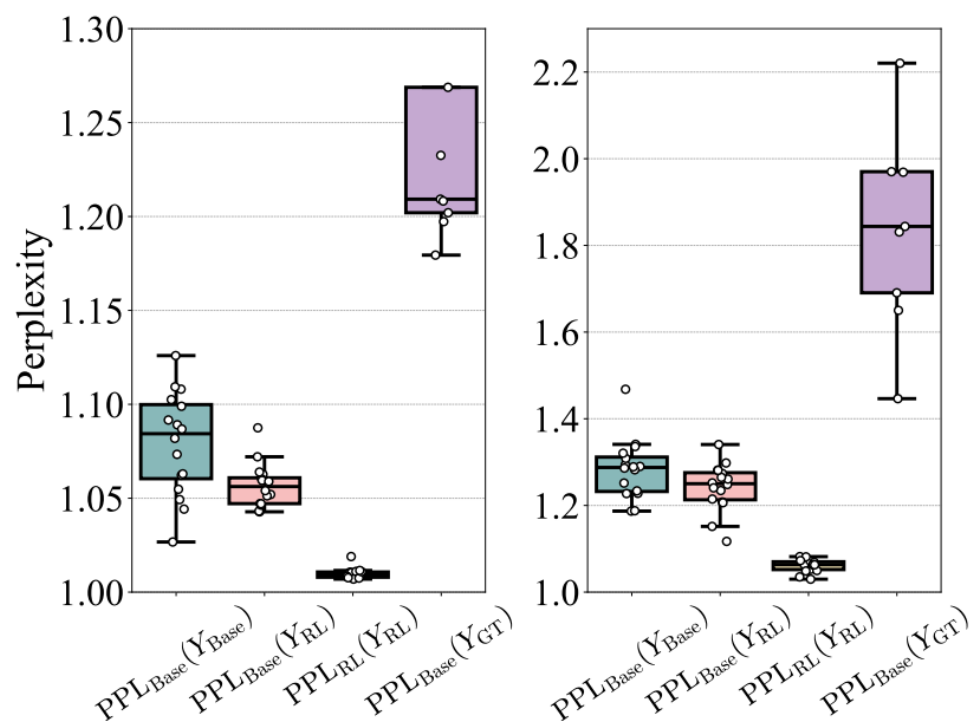
Base	SimpleRLZoo	AIME24	MATH500
✓	✓	63.3%	92.4%
✓	✗	13.3%	3.6%
✗	✓	0.0%	1.0%
✗	✗	23.3%	3.0%

1. Many cases where the base model solves a problem but the RLVR model fails;
2. Very few where RLVR succeeds while the base model does not

Key findings

Perplexity Analysis.

(randomly sample two problems from AIIME24 and generate 16 responses for each problem)



$$\text{PPL}_m(\mathbf{Y} | x) = \exp \left(-\frac{1}{T} \sum_{t=1}^T \log P(y_t | x, y_1, \dots, y_{t-1}) \right),$$

1. The responses from RL-trained models are highly likely to be generated by the base model.
2. RLVR mainly sharpens the distribution within the base model's prior rather than expanding beyond it.

Key findings

Implementation

Instruction tuning: using short instruction-response pairs

Distillation: the training data consist of long CoT reasoning traces generated by the teacher model.

Conclusion

Distillation transfers new patterns from teacher models, expanding reasoning boundaries (pass@k consistently above base models).

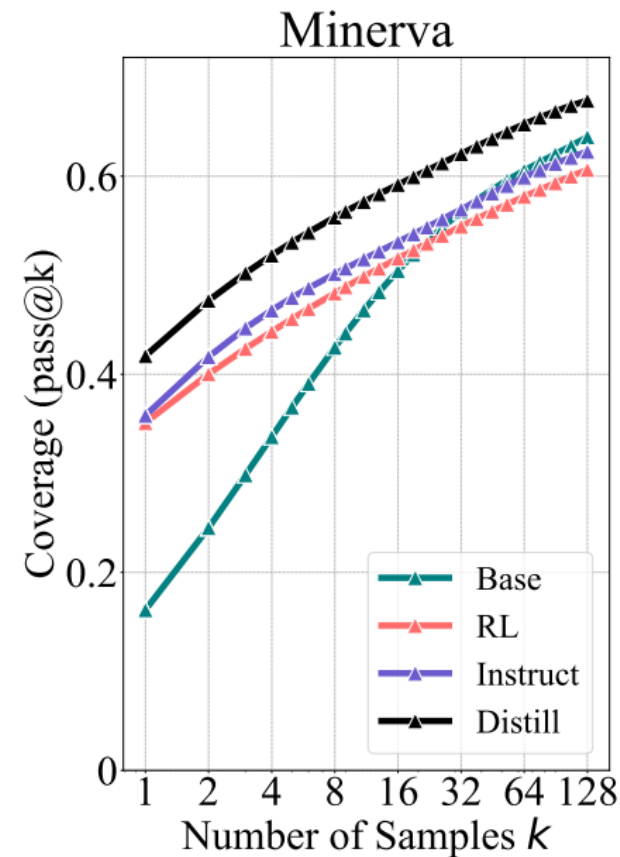


Figure 7: pass@ k of base, Instruct, RLVR, and distilled models.

Tiny Model, Big Logic: Diversity-Driven Optimization Elicits Large-Model Reasoning Ability in **VibeThinker-1.5B**

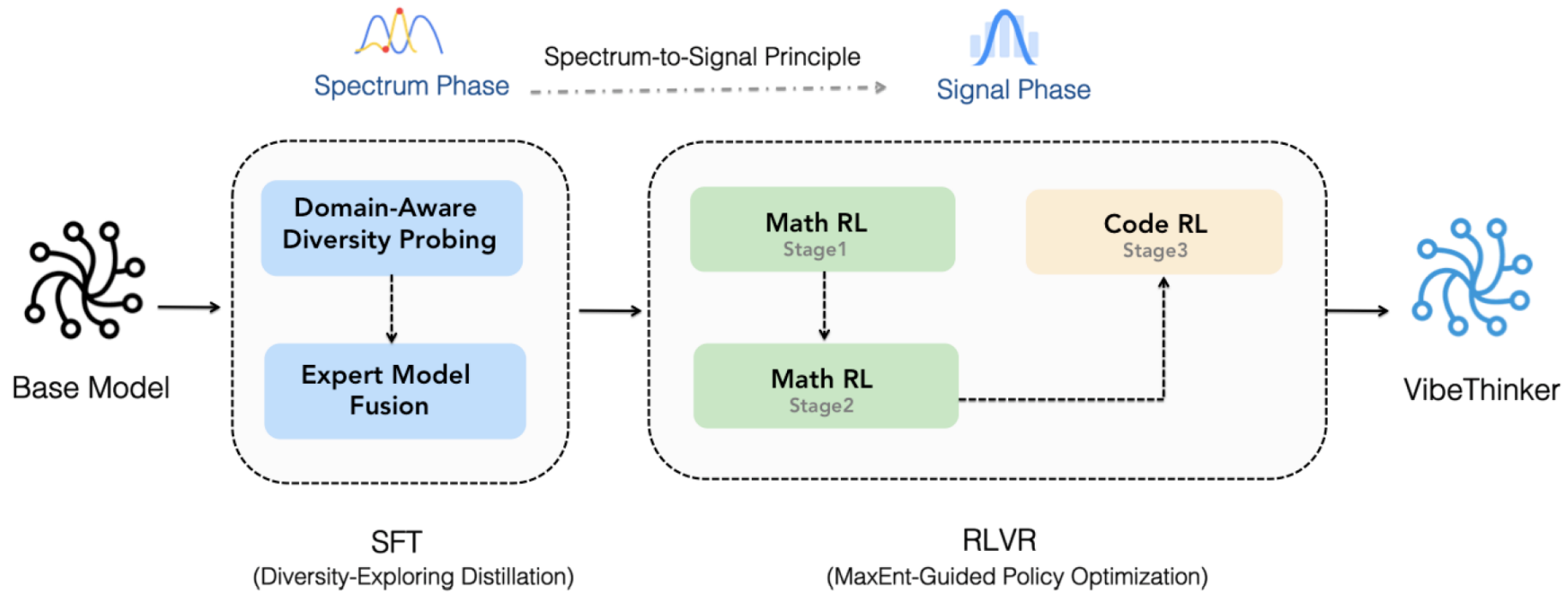
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Background

- Context:** RLVR brings Long CoT for LLMs, marking significant progress in reasoning technology.
- Common Belief:** Reasoning ability = Parameter scaling (e.g., DeepSeek R1: 671B; Kimi K2: >1T); Existing tiny models (1.5B-3B) fail to fully unlock logical reasoning potential.
- Core Question:** Can a 1.5B-parameter "Tiny Model" match state-of-the-art large models in reasoning—without extreme parameter scaling?

Framework: Spectrum-to-Signal Principle



The Spectrum Phase (SFT): The primary goal of SFT is not to converge on a single optimal answer, but to generate a rich and diverse "spectrum" of plausible solutions.

The Signal Phase (RL): The role of RL is then to identify and amplify the correct "signal" from within this pre-established spectrum.

SFT Stage (Spectrum Phase)

Two-Stage Diversity-Exploring Distillation

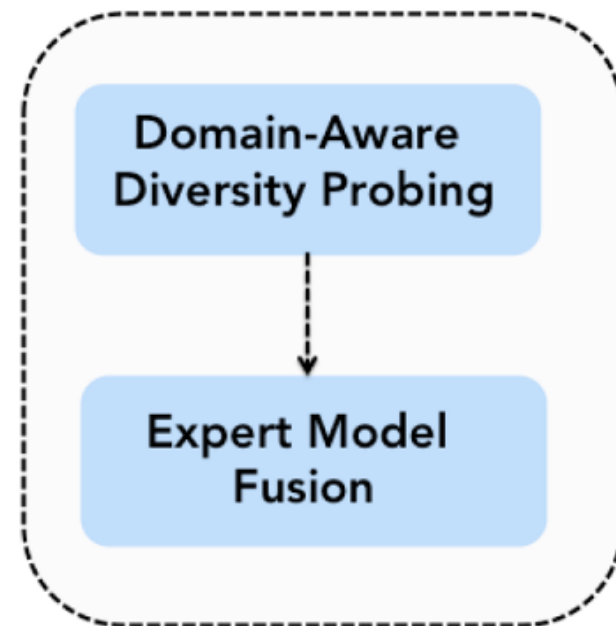
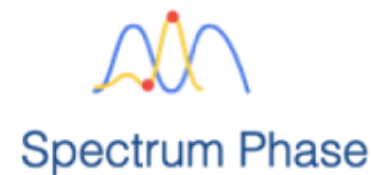
1.Domain-Aware Diversity Probing

1. Split tasks (e.g., math) into subdomains (algebra, geometry, calculus, statistics).
2. Select "expert models" with the highest Pass@K for each subdomain.

2.Expert Model Fusion

1. Weighted parameter averaging (equal weights for VibeThinker-1.5B).

$$\mathbf{M}_{\text{Merge}}^{\text{SFT}} = \sum_{i=1}^N w_i M_i^*$$



SFT
(Diversity-Exploring Distillation)

RL Stage (Signal Phase)

MaxEnt-Guided Policy Optimization (MGPO)

Core Logic: Prioritize training on high-uncertainty problems (accuracy $\approx 50\%$)

$$D_{\text{ME}}(p_c(q) \| p_0) = p_c(q) \log \frac{p_c(q)}{p_0} + (1 - p_c(q)) \log \frac{1 - p_c(q)}{1 - p_0}$$

$$w_{\text{ME}}(p_c(q)) = \exp(-\lambda \cdot D_{\text{ME}}(p_c(q) \| p_0)), \quad \text{where } p_0 = 0.5, \lambda \geq 0 \quad p_c(q) = \frac{1}{G} \sum_{i=1}^G \mathbb{I}(r_i = 1)$$

$$\mathcal{A}'_j(q) = w_{\text{ME}}(p_c(q)) \cdot \mathcal{A}_j(q)$$

$$\mathcal{J}_{\text{MGPO}}(\theta) = \mathbb{E}_{(q,y) \sim \mathcal{D}} \left[\mathbb{E}_{\{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot | q)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} (\min(r_{i,t}(\theta) \mathcal{A}'_{i,t}(q), \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \mathcal{A}'_{i,t}(q))) \right] \right]$$

Experiments

Table 1. Comparison of Post-Training Costs

Models	Size	AIME25 Score	GPU Type	GPU Hours	Total Cost
DeepScaleR	1.5B	31.5	A100	3.8K	\$4.5K
MiniMax-M1	456B	74.6	H800	258K	\$535K
DeepSeek-R1	671B	70.0	H800	147K	\$294K
VibeThinker	1.5B	74.4	H800	3.9K	\$7.8K

Table 2. Performance of VibeThinker-1.5B on Core Benchmarks (Small Reasoning Models)

Model			Mathematics				Coding		Knowledge
Name	Params	Institution	AIME 2024	AIME 2025	MATH 500	HMMT25	LCB v5	LCB v6	GPQA Diamond
L1-Max	1.5B	CMU	27.7	21.0	84.7	9.9	–	–	–
STILL-3	1.5B	RUC	34.7	24.0	86.6	13.9	–	–	–
DeepScaleR	1.5B	UC Berkeley	43.1	31.5	87.8	19.0	–	16.3	20.5
DeepSeek-Distill-Qwen	1.5B	DeepSeek	28.5	22.7	82.9	13.6	16.8	12.8	15.9
FastCURL-v3	1.5B	Tencent	49.6	34.4	90.5	21.5	–	20.6	23.2
ProRL	1.5B	NVIDIA	48.1	33.3	91.9	20.5	23.8	–	41.8
Hunyuan-0729	1.8B	Tencent	56.7	–	86.0	–	31.5	–	35.8
Qwen3	1.7B	Alibaba	48.3	36.8	93.4	23.6	33.2	26.9	40.1
SmolLM 3	3B	HuggingFace	–	36.7	87.5	26.0	27.6	29.1	41.7
Base Model	1.5B	–	6.7	4.3	58.5	0.6	0.0	0.0	16.4
VibeThinker-1.5B	1.5B	Weibo	80.3	74.4	95.0	50.4	55.9	51.1	46.7

Experiments

